



Student Performance Analytics

Enterprise Data Portfolio & End-to-End Business
Intelligence Report

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Tech Stack: Python, SQL, Power BI, Excel, Git

Repository: github.com/moneshr/student-performance-analytics

Date: August 2026

PRODUCTION READY

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01 Executive Summary

The **Student Performance Analytics** initiative is an end-to-end data engineering and business intelligence solution designed to track, evaluate, and forecast academic success across secondary and higher-secondary educational frameworks. Managing an active cohort of **1,250 students**, this enterprise solution aggregates multi-source student metrics to provide institutional stakeholders with real-time visibility into academic trends.



Deep-dive segment analysis revealed 58 students currently at risk (scoring below 40%), prompting targeted interventions. By deploying an automated data pipeline and a 5-page interactive Power BI dashboard, leadership can now proactively deploy pedagogical resources, optimize subject curricula, and secure better graduation outcomes.

KEY INSIGHT

Monthly trend lines indicate that as average attendance climbed steadily from 72% in January to 90% in December, institutional average scores concurrently rose from 62% to 86% — proving that regular classroom engagement is a strong driver of academic success.

02 Project Overview

Educational institutions generate vast amounts of disparate administrative, behavioral, and scoring records. Without a centralized analytics layer, tracking student progress remains reactive. This project bridges that gap by establishing a professional, production-grade analytics repository.

The system moves raw data — extracted via Python from transactional databases — through a cleaned SQL repository, feeding directly into a polished, executive-ready Power BI presentation layer. It tracks foundational metrics such as attendance trends, gender distributions, subject-wise proficiencies, and cumulative grade distributions across academic sessions (e.g., 2023–2024).

03 Business Problem

School administrators, academic heads, and educators face several operational hurdles:

Challenge	Impact
Late Identification of At-Risk Students	Instructors often discover failing grades only after final examinations, preventing timely remediation.
Siloed Performance Metrics	Attendance records, subject scores, and demographic files exist in separate data islands, obscuring root-cause correlations between attendance and performance.
Lack of Longitudinal Tracking	Difficulty in visualizing monthly score trajectories and comparing current metrics against previous academic periods.
Manual Reporting Overhead	Administrative teams waste hours compiling static Excel reports rather than focusing on strategic student counseling.

04 Objectives

- **Centralize Data Operations** — Build a unified SQL data warehouse schema fed by robust Python extraction scripts.
- **Automate Insight Generation** — Deliver an interactive, multi-view Power BI dashboard reflecting real-time updates and granular filters (Academic Year, Class, Section, Subject, and Date Range).
- **Mitigate Academic Failure** — Isolate the 58 at-risk students and map intervention paths based on attendance drops and grade flags.
- **Optimize Curriculum Strategy** — Highlight high-performing departments (e.g., Computer Science at 85.70%) vs. improvement zones to balance institutional focus.

05 Technology Stack

The architecture integrates industry-standard tools to guarantee scalability, security, and reproducibility:

Python (Pandas / NumPy)

SQL (PostgreSQL / SQLite)

Power BI Enterprise

DAX & Power Query

Excel / CSV Templates

Git & GitHub Version Control

06Dataset Description & Data Dictionary

The analysis is anchored on `student_performance.csv`, capturing structured records for **1,250 students**.

Field Name	Data Type	Description
StudentID	Integer	Unique identifier assigned to each student.
StudentName	Varchar	Full name of the student.
Gender	Varchar	Gender classification (Male, Female).
AcademicYear	Varchar	Active academic cycle (e.g., 2023–2024).
Class & Section	Varchar	Enrolled grade level and section identifier.
Subject	Varchar	Specific subject domain (Mathematics, Science, English, Social Studies, Computer).
AttendanceRate	Float	Cumulative attendance percentage over the active term.
AverageScore	Float	Calculated mean score across evaluations (0–100 scale).
Grade	Char	Letter grade tier (A, B, C, D, F).
Status	Varchar	Institutional standing (Pass or Fail).

07 Data Pipeline Architecture

[Raw CSV Sources] → Python Data Cleaning & Validation (Pandas/NumPy) → [Staging Area]
→ SQL Transformations & Relational Modeling → [Enterprise Data Warehouse]
→ Direct Query / Import Mode → [Power BI Enterprise Dashboard Suite]

Stage	Description
1. Ingestion	Raw files (student_performance.csv) are ingested into the pipeline.
2. Processing	Python scripts execute data type casting, outlier detection, and null-value imputation.
3. Storage	Cleaned relations are written to a structured SQL database schema optimized for reporting joins.
4. Presentation	Power BI connects to the warehouse using optimized DAX measures to drive sub-second rendering across all visual widgets.

08 Dashboard Overview & Visualizations

The Power BI solution is structured into cohesive thematic pages mirroring modern enterprise design standards (Dark Mode Executive Theme with high-contrast indicator cards).

Figure 8.1 — Power BI Global Dashboard Hub

Top navigation slicers, KPI cards, and trend analytics.

Figure 8.2 — Dashboard Overview Screen

Macro-level health check, grade distributions, and top performers.

Figure 8.3 — Student Overview Module

Demographic breakdown, gender distribution, and attendance ranges.

Figure 8.4 — Performance Analysis Suite

Subject spider chart, at-risk trackers, and performance extremes.

KPI Metric	Value	Definition & Calculation
Total Students	1,250	Active registry count filtered by academic year boundaries.
Average Score	78.45%	Mean percentage score across all evaluated assignments and exams.
Pass Percentage	92.40%	Proportion of students securing scores $\geq$ 40%.
Average Attendance	88.32%	Mean presence metric, highly correlated with score success trajectories.
Students At Risk	58	Operational threshold tracker identifying individuals with scores < 40%.

- **Attendance-Performance Correlation** — Monthly trend lines indicate that as average attendance climbed steadily from 72% in January to 90% in December, institutional average scores concurrently rose from 62% to 86%.
- **Subject Proficiency Disparities** — Computer Science leads department averages at 85.70%, whereas Social Studies registers the lowest foundational mean at 74.21%, indicating a need for pedagogical review in humanities.
- **Gender Parity** — Academic distributions remain exceptionally balanced, with male students comprising 52.0% and female students 48.0%, showing equitable access and performance parity across sections.
- **Tail-End Vulnerability** — While overall pass rates sit comfortably at 92.40%, the absolute count of 58 at-risk students requires targeted tracking to prevent dropouts.

11 Strategic Recommendations

1. **Mandatory Attendance Intervention Triggers** — Implement an automated alert system whenever a student's monthly attendance dips below 75%, bridging the gap before final grade drops occur.
2. **Curriculum Peer-Mentoring** — Leverage top-tier students (such as top performer Rohit Sharma, 98.20%) within peer-tutoring cohorts to lift struggling segments in Social Studies and English.
3. **Resource Realignment** — Allocate additional academic support hours and digital learning tooling toward the 58 identified at-risk students to shrink the failure rate below 5%.

12 Future Enhancements

- **Predictive Machine Learning Integration** — Embed scikit-learn classification models directly into the Python pipeline to predict student dropout risk prior to term exams.
- **Automated Email Alerting** — Configure automated Power Automate hooks to dispatch weekly progress reports directly to homeroom teachers.
- **Cloud Migration** — Transition the local SQL database instance to a cloud-managed warehouse (Snowflake / Azure SQL) for real-time multi-campus scalability.

13 Repository Structure

```
student-performance-analytics/  
|  
├─ data/  
|   ├─ raw/                student_performance.csv  
|   └─ processed/          cleaned_student_data.sql  
|  
├─ notebooks/  
|   └─ 01_exploratory_data_analysis.ipynb  
|  
├─ scripts/  
|   ├─ data_pipeline.py  
|   └─ validation_checks.py  
|  
├─ dashboard/  
|   ├─ student_performance_model.pbix  
|   ├─ powerbi-dashboard_2.png  
|   ├─ dashboard-overview_2.png  
|   ├─ student-overview_2.png  
|   └─ performance-analysis_2.png  
|  
├─ README.md  
└─ requirements.txt
```

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"Dedicated to transforming complex educational and business datasets into actionable strategic intelligence."

GitHub Repository: github.com/moneshr/student-performance-analytics